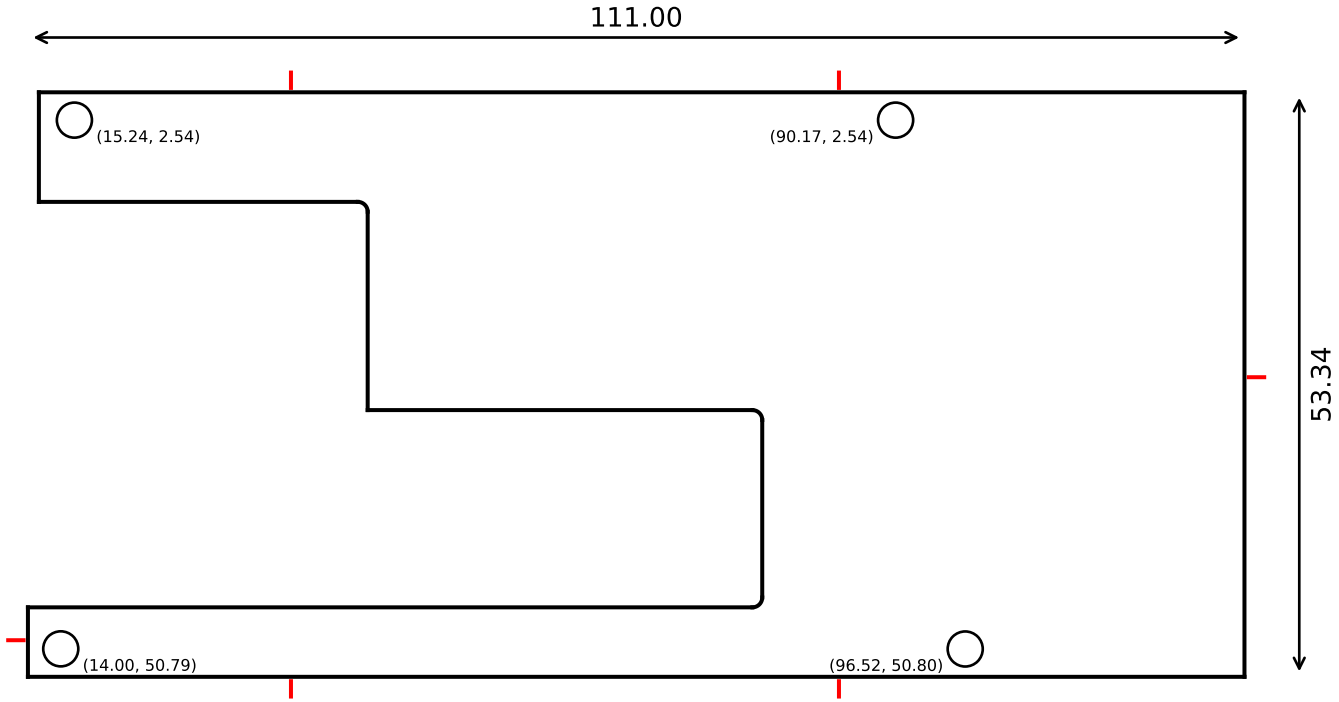
giga-adapter — fabrication drawing



Finished thickness: 1.6 mm +/-10%

4 layers, FR-4

35 um copper, all layers

ENIG

Green mask / white silk

IPC Class 2

Electrical test: 100%

Red: bridge suggestions

Layer order: F.Cu / In1.Cu / In2.Cu / B.Cu. Routed Edge.Cuts and drill files are authoritative. Four-layer dielectrics: 0.14 / 1.20 / 0.14 mm, nominal finished 1.6 mm. GIGA concave milling radii: 0.9 mm.

Minimum track 0.15 mm; copper clearance 0.150 mm. Through via drill 0.30 mm, diameter 0.65 mm, annular ring >=0.15 mm. Exposed vias, no via-in-pad. Zero source mask expansion; processed fine-pitch webs >=0.10 mm. Component-hole rings >=0.30 mm.

AISLER bare boards only; ENIG on both boards. Use the ODB++ upload ZIP, including the aisler-bridges document layer. Keep the closed profile unchanged. PTH and NPTH drills are separate. No assembly or stencil is ordered.

Origin in Gerbers, drills and placement: X=0, Y=lower board edge; positive Y upward. Drawing coordinates above use KiCad top-left Y-down for mechanical reference.

Optional paste reference only (no stencil ordered): 0.10 mm at Molex 505110, with 100% land-area openings; 0.10–0.12 mm at 0402/TSSOP parts. DF23 paste follows its separate manufacturer drawing. No substitutions for connector suffixes or header tail lengths.

giga-adapter — assembly instructions

Use assembly-F.pdf and assembly-B.pdf for reference designators, pin numbers and component orientation. Bottom drawing is mirrored to show the actual bottom view. Native KiCad placement rotations are retained in placement.csv.

All components in bom-all.csv are FIT and installed separately by the builder. bom-smd.csv and placement.csv are placement aids only. HAND denotes through-hole installation; no assembly service is included.

Inspect every 0.5 mm connector contact for bridges, insufficient solder and coplanarity. Check soldermask webs and that no solder or exposed via intrudes into the DF23 contact-band keepout.

J1 Molex 505110 is TOP mounted, cable mouth facing the right board edge. Pin 1 is at the lower end when the board is viewed from the top. Check the silkscreen 1 marking and the numbered assembly drawing.

J10 DC-005: HAND solder, 5.5/2.1mm barrel opening faces left into the service cutout. CENTRE POSITIVE 24V: pin 1 = +24V, pins 2/3 = GND. Three plated slots: width 1.2mm, lengths 4.0/3.6/4.0mm. U8 TSR 1-24120 is also HAND soldered.

U7 thermal pad and RTN belong to their own copper island; never solder-bridge them to GND. D1 is bidirectional; D2/D3 cathodes follow their band and pad 1.

J2–J9 stacking headers are HAND soldered, sockets above the PCB and tails below. J2–J8: Samtec SSQ -03, 10 mm tails. J9: SSQ-112-04-G-D, 14.83 mm tails. Verify exact suffixes in the full BOM.

GIGA top to carrier underside: 13.0 mm. Lower two mounting positions use M2 insulating support hardware; upper holes are alignment clearances without screw heads. Support the cable end independently.

Do not bridge isolated pass-through contacts to adapter ground or power. Motor/EtherCAT contacts remain accessible through the stacking sockets and are electrically unconnected on this PCB.

First-board power, timing, touch and supply-sequencing tests are listed in README.md. They have not been performed.